
Assignment 2

CS771: Machine Learning

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Part 1: Mathematical Derivation

We want to learn the Maximum Likelihood Estimation (MLE) for the 17-bit PUF model characterized by parameters $(\mathbf{w}, b) \in \mathbb{R}^{17} \times \mathbb{R}$. We are given the 16-bit challenges $c_i \in \{0, 1\}^{16}$ and responses $r_i \in \{0, 1\}$, with latent variables $z_i \in \{0, 1\}$ representing the hidden middle bits.

By applying the chain rule of probability, the joint distribution of observing the response and the latent variable is:

$$P[r_i, z_i | c_i, \mathbf{w}, b] = P[r_i | z_i, c_i, \mathbf{w}, b] P[z_i | c_i, \mathbf{w}, b] \quad (1)$$

We are given the latent prior $P[z_i | c_i, \mathbf{w}, b] = 0.5$ for all $i \in [8000]$. Applying Melbo's logistic likelihood model, the joint probability becomes:

$$P[r_i, z_i | c_i, \mathbf{w}, b] = 0.5 \cdot \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) \quad (2)$$

Taking the negative log-likelihood over all 8000 independent CRPs and utilizing the winner-take-all approximation (Hard EM) yields our objective function:

$$\begin{aligned} \min_{\mathbf{w}, b, \{z_i\}} - \sum_{i=1}^{8000} \ln(P[r_i, z_i | c_i, \mathbf{w}, b]) &= \min_{\mathbf{w}, b, \{z_i\}} \sum_{i=1}^{8000} \left[-\ln(0.5) - \ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) \right] \\ &= \min_{\mathbf{w}, b, \{z_i\}} \sum_{i=1}^{8000} -\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) \end{aligned} \quad (3)$$

Note that the $-\ln(0.5)$ term is a constant and vanishes from the optimization objective. Because this objective is non-convex with respect to simultaneously finding $(\mathbf{w}, b)$ and $\{z_i\}$, we utilize an alternating optimization algorithm consisting of two steps.

1. The M-Step (Update parameters given latent variables)

Assume the current estimates of the latent variables $\{z_i\}$ are fixed. The insertion function $I(c_i, z_i)$ produces fully known 17-bit feature vectors. We optimize the parameters $(\mathbf{w}, b)$ by minimizing the standard binary cross-entropy loss:

$$(\mathbf{w}, b) = \arg \min_{\mathbf{w} \in \mathbb{R}^{17}, b \in \mathbb{R}} \sum_{i=1}^{8000} -\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) \quad (4)$$

This is a convex optimization problem that can be solved using standard logistic regression solvers.

2. The E-Step (Update latent variables given parameters)

Assume the model parameters $(\mathbf{w}, b)$ are fixed. Because the CRPs are independent, the global summation decouples. For each sample i , we find the optimal latent bit $z_i \in \{0, 1\}$ that minimizes its individual negative log-likelihood:

$$z_i = \arg \min_{z \in \{0, 1\}} -\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z)) + b) \right) \quad (5)$$

Since the negative logarithm and the sigmoid function are strictly monotonic, minimizing the negative log-sigmoid is mathematically equivalent to maximizing the linear margin. Therefore, the update rule simplifies to:

$$z_i = \arg \max_{z \in \{0, 1\}} \left[(2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z)) + b) \right] \quad (6)$$

By iterating between Step 1 and Step 2 until the assignments for $\{z_i\}$ converge, we approximate the Maximum Likelihood Estimation for the PUF model.

Part 3: Updated Latent Variable MLE Derivation

We are given that the latent variables z_i are not random, but are generated by a hidden 16-bit PUF parameterized by $(\mathbf{u}, a) \in \mathbb{R}^{16} \times \mathbb{R}$. We must jointly learn this hidden model alongside the 17-bit PUF parameterized by $(\mathbf{w}, b) \in \mathbb{R}^{17} \times \mathbb{R}$.

The joint distribution of the final response and the latent variable is given by the product of the updated likelihood and the new latent prior:

$$P[r_i, z_i | c_i, \mathbf{w}, b, \mathbf{u}, a] = P[r_i | z_i, c_i, \mathbf{w}, b] P[z_i | c_i, \mathbf{u}, a] \quad (7)$$

Applying the winner-take-all (Hard EM) approximation, maximizing the marginal likelihood is mathematically equivalent to minimizing the negative joint log-likelihood over all parameters and hard latent assignments $\{z_i\}$:

$$\begin{aligned} & \min_{\mathbf{w}, b, \mathbf{u}, a, \{z_i\}} \sum_{i=1}^{8000} -\ln P[r_i, z_i | c_i, \mathbf{w}, b, \mathbf{u}, a] \\ &= \min_{\mathbf{w}, b, \mathbf{u}, a, \{z_i\}} \sum_{i=1}^{8000} \left[-\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) - \ln \sigma \left((2z_i - 1)(\mathbf{u}^\top \phi(c_i) + a) \right) \right] \end{aligned} \quad (8)$$

Because this joint problem is non-convex, we apply an alternating optimization algorithm:

1. The M-Step (Update models given latent variables)

Assume the current estimates of the latent variables $\{z_i\}$ are fixed. The joint optimization objective cleanly decouples into two independent convex problems, corresponding to standard logistic regression losses:

$$(\mathbf{w}, b) = \arg \min_{\mathbf{w}, b} \sum_{i=1}^{8000} -\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z_i)) + b) \right) \quad (9)$$

$$(\mathbf{u}, a) = \arg \min_{\mathbf{u}, a} \sum_{i=1}^{8000} -\ln \sigma \left((2z_i - 1)(\mathbf{u}^\top \phi(c_i) + a) \right) \quad (10)$$

2. The E-Step (Update latent variables given models)

Assume the parameters $(\mathbf{w}, b)$ and $(\mathbf{u}, a)$ are fixed. Because the samples are independent, we

evaluate each sample i individually. We find the optimal bit $z_i \in \{0, 1\}$ that minimizes the sum of the negative log-likelihoods (which is equivalent to maximizing the combined log-likelihood score):

$$z_i = \arg \max_{z \in \{0, 1\}} \left[\ln \sigma \left((2r_i - 1)(\mathbf{w}^\top \phi(I(c_i, z)) + b) \right) + \ln \sigma \left((2z - 1)(\mathbf{u}^\top \phi(c_i) + a) \right) \right] \quad (11)$$

By continuously alternating between the M-Step and E-Step until $\{z_i\}$ stabilizes, we force the latent variables to satisfy the constraints of both models simultaneously, arriving at our updated Maximum Likelihood Estimation.

Part 5: Model Comparison and Analysis

Based on the evaluation of both algorithms, the outputs of `my_latent()` and `my_latent_updated()` strongly **disagree**.

- (a) **Did you observe $w \approx \tilde{w}$?**

No, the 17-dimensional weight vectors did not match.

- (b) **Did you observe $b \approx \tilde{b}$?**

No, the bias terms were significantly different between the two models.

- (c) **Agreement of Latent Variables:**

I observed that the condition $2 \cdot z_i - 1 = \text{sign}(\tilde{u}^\top \phi(c_i) + \tilde{a})$ held true for exactly **0.5238 (52.38%)** of the 8000 training CRPs. Since the variables are binary $\{0, 1\}$, this 50% agreement is effectively equivalent to random chance.

Explanation of Observations:

The fundamental disagreement stems from the different constraints placed on the latent variables during optimization.

In `my_latent()`, we assume a flat prior ($P = 0.5$). The latent variables z_i are entirely unconstrained, giving the algorithm 8000 "free choices." As a result, the algorithm simply memorizes the dataset; it arbitrarily flips the latent bits to whichever value artificially forces the 17-bit model to perfectly predict the training responses. This leads to an overfitted model (w, b) that hallucinates the middle bits to minimize training error, completely missing the true hardware parameters.

In contrast, `my_latent_updated()` introduces the 16-bit PUF as a generative prior for the latent variables. A middle bit cannot be chosen simply because it minimizes the 17-bit model's error; it must simultaneously be a highly probable output of the 16-bit physical system $(\tilde{u}, \tilde{a})$. This joint dependency acts as a powerful regularizer. It prevents the 17-bit model from cheating and forces the EM algorithm to converge on the true underlying physical parameters of both systems. Because Algorithm 1's variables were chosen to overfit rather than reflect physics, they only match Algorithm 2's physically-grounded predictions at a rate equal to random chance.